Combinatorics: Identities, Generating Functions, Catalan Numbers, and More

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§1.1 Binomial identities

The note opens with several binomial identities and, more importantly, their combinatorial meanings. The basic identity

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n$$

counts all subsets of an n-element set: choosing a subset means choosing its size i and then choosing i elements.

Pascal's identity

$$\binom{n}{i} + \binom{n}{i+1} = \binom{n+1}{i+1}$$

can be proved by fixing a distinguished element in an $(n+1)$-element set. An $(i+1)$-subset either contains the distinguished element or it does not. The two cases give the two terms on the left.

Vandermonde's identity

$$\sum_{i} \binom{m}{i} \binom{n}{r-i} = \binom{m+n}{r}$$

counts choosing r balls from two boxes with m and n balls. If i balls are chosen from the first box, then $r-i$ are chosen from the second.

Another useful identity is

$$\binom{n}{r}\binom{n-r}{k-r} = \binom{n}{k}\binom{k}{r}.$$

Both sides count a pair (R, K) with $R \subseteq K \subseteq [n]$, $|R| = r$, and $|K| = k$.

§1.2 Good subsets

A nonempty subset $A \subseteq \{1, 2, \ldots, n\}$ is called good if

$$|A| \leq \min_{x \in A} x.$$

Let a_n be the number of good subsets. The problem asks one to prove

$$a_{n+2} = a_{n+1} + a_n + 1.$$

A clean proof splits good subsets of $[n+2]$ into cases.

First, if $n+2 \notin A$, then A is just a good subset of $[n+1]$, giving a_{n+1} possibilities.

Second, if $A = \{n+2\}$, this gives one extra subset.

Third, suppose $n+2 \in A$ and $|A| \geq 2$. Remove $n+2$ and subtract 1 from every remaining element. If

$$A' = A \setminus \{n+2\},$$

then the condition becomes

$$|A'| + 1 \leq \min A'.$$

After shifting $B = \{x - 1 : x \in A'\}$, this is exactly

$$|B| \leq \min B,$$

so B is a good subset of $[n]$. This gives a_n possibilities. Hence

$$a_{n+2} = a_{n+1} + a_n + 1.$$

A summation proof fixes $|A| = k$. Every element must then be at least k, so the remaining data are chosen from a tail interval, and repeated Pascal identities yield the recurrence.

§1.3 Generating functions

For a sequence $(a_n)_{n\geq 0}$, its ordinary generating function is

$$f(x) = \sum_{n\geq 0} a_n x^n.$$

The note emphasizes a key principle: coefficients can encode counting problems.

For example, the number of nonnegative integer solutions of

$$x_1 + x_2 + \cdots + x_k = n$$

is the coefficient of x^n in

$$(1 + x + x^2 + \cdots)^k = \frac{1}{(1-x)^k}.$$

Using the binomial expansion,

$$\frac{1}{(1-x)^k} = \sum_{n\geq 0} \binom{n+k-1}{k-1} x^n,$$

so the number of solutions is

$$\binom{n+k-1}{k-1}.$$

The notes also use generating functions to separate parity or color constraints. A typical trick is to multiply several series, each recording the allowed choices for one box or color, and then read off the coefficient of the target power.

§1.4 A probability example with repeated trials

For repeated trials, a generating function rewrites nested binomial sums using exponentials. The important method is not the final closed form alone but the transformation:

$$\text{complicated convolution} \quad \longrightarrow \quad \text{product/composition of generating functions.}$$

This is a standard way to turn repeated independent operations into algebra. Binomial coefficients record choices; exponential series record independent counts; coefficient extraction recovers the desired probability.

§1.5 Catalan numbers

The note then turns to Catalan numbers. One model is the number of ways to triangulate a convex $(n+2)$-gon:

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

For example, a hexagon has 14 triangulations.

Let h_n denote the number of triangulations of an $(n+2)$-gon. Choose the triangle incident with a fixed side. If its third vertex cuts the polygon into two smaller polygons, the numbers of triangulations multiply. Therefore

$$h_n = \sum_{i=0}^{n-1} h_i h_{n-1-i}, \qquad h_0 = 1.$$

If

$$H(x) = \sum_{n\geq 0} h_n x^n,$$

then the recurrence gives

$$H(x) = 1 + xH(x)^2.$$

Solving the quadratic and choosing the branch with $H(0) = 1$ gives

$$H(x) = \frac{1-\sqrt{1-4x}}{2x}, \qquad h_n = C_n.$$

This derivation explains why Catalan numbers appear in triangulations, parentheses, Dyck paths, and many noncrossing structures: all of them are recursively split into a left and a right part.

§1.6 Convex sets and convex hulls

The geometry part defines a convex set S by the condition that for any $x, y \in S$ and any $t \in [0,1]$,

$$tx + (1-t)y \in S.$$

Equivalently, S contains the line segment between any two of its points. The note contrasts convex and non-convex examples.

For finitely many points $x_1, \ldots, x_k$, their convex hull is

$$\left\{ \sum_{i=1}^{k} t_i x_i : \ t_i \geq 0, \ \sum_{i=1}^{k} t_i = 1 \right\}.$$

Induction on k proves the claim: the case $k = 2$ is the definition of a segment, and the induction step writes the convex combination as a combination of one point with a convex combination of the remaining points.

§1.7 Separating convex sets

The note sketches a separation idea. For two disjoint convex sets, one tries to find a point pair realizing minimal distance and then uses the perpendicular bisector of the segment between them as a separating line. In symbols, if $u \in S_1$ and $v \in S_2$ are closest points, then the hyperplane perpendicular to $v - u$ near the midpoint separates the two sets.

In finite-dimensional Euclidean geometry, this is the intuitive core of the separating hyperplane theorem. Projection onto a line produces inequalities involving squared distances, and the algebra expands

$$\|u + t(a - u) - v\|^2$$

for small t and extracts a sign condition by letting $t \to 0$.

§1.8 Planar graphs and Euler-type counting

Graph theory and planar configurations provide another counting mechanism: a planar drawing constrains vertices, edges, and faces through Euler-type formulas.

This connects naturally with the previous convex-geometry material: triangulations, planar graphs, polygon decompositions, and convex hulls are different languages for controlling incidence data.

§1.9 Graphs and adjacency matrices

A graph G consists of a vertex set $V = \{v_1, \ldots, v_n\}$ and an edge relation. For a simple graph, the adjacency matrix $A(G) = (a_{ij})$ is defined by

$$a_{ij} = \begin{cases} 1, & \text{if there is an edge from } v_i \text{ to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

For a multigraph, a_{ij} records the number of edges from v_i to v_j.

The source quotes the standard theorem: the (i, j)-entry of $A(G)^\ell$ equals the number of walks of length ℓ from v_i to v_j. The proof is matrix multiplication. For $\ell = 2$,

$$(A^2)_{ij} = \sum_k a_{ik} a_{kj},$$

which counts all intermediate vertices v_k. The general case follows by induction.

§1.10 The deeper habit

Several counting languages share one habit: replace a raw problem by a structure-preserving encoding. Subsets become binomial coefficients, recursions become generating functions, triangulations become Catalan recurrences, convex hulls become linear combinations, and walks become matrix powers.

§1.11 Generating functions as algebraic counting

A generating function packages a sequence into a formal series. Products encode decomposition of objects, and coefficient extraction reads off counts:

$$[x^n]A(x)B(x) = \sum_{k=0}^{n} a_k b_{n-k}.$$

§1.12 Catalan structures

Catalan numbers count many recursively decomposable objects:

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Their generating function satisfies

$$C(x) = 1 + xC(x)^2.$$

The quadratic equation records the root-subtree decomposition.

§1.13 Convex hulls and separation

The separating hyperplane theorem says disjoint convex sets can often be separated by a linear functional. This is the geometric dual of optimization: a linear functional witnesses non-intersection.

§1.14 Generating functions, Catalan recursion, and spectra

Binomial identities, generating functions, Catalan numbers, probability trials, and convex separation all share one habit: encode the structure so that algebraic or linear operations reveal the count or obstruction.

§1.15 Catalan numbers as recursive geometry

Catalan numbers count many objects: correctly matched parentheses, triangulations of polygons, rooted binary trees, Dyck paths, and noncrossing partitions. A standard formula is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

The recursion

$$C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$$

comes from choosing the first return of a Dyck path or the root split of a binary tree.

The generating function $C(x) = \sum_{n\geq 0} C_n x^n$ satisfies

$$C(x) = 1 + xC(x)^2.$$

Solving gives

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}.$$

Thus an elementary recursion becomes an algebraic generating function.

§1.16 Lattice paths and reflection principle

Many Catalan formulas are proved by the reflection principle. The number of paths from $(0,0)$ to (n,n) that never go above the diagonal is

$$\binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n}.$$

The subtracted term counts bad paths by reflecting the part up to the first diagonal crossing.

This is a model example of an involutive or bijective proof. Rather than evaluating a sum, one constructs a symmetry pairing bad objects with easier objects.

§1.17 Graph walks and adjacency matrices

If G is a finite graph with adjacency matrix A, then

$$(A^k)_{ij}$$

counts walks of length k from vertex i to vertex j. This turns walk counting into matrix powers.

If A is diagonalizable with eigenvalues $\lambda_1, \ldots, \lambda_n$, the growth of walks is controlled by the largest eigenvalues. In a regular connected graph, the largest eigenvalue is the degree, and the second-largest eigenvalue controls expansion and mixing.

Thus an elementary graph-walk count leads directly to spectral graph theory.

§1.18 Random walks and Markov chains

Normalize the adjacency matrix of a d-regular graph by

$$P = \frac{1}{d}A.$$

Then P is the transition matrix of a random walk. The distribution after k steps is

$$\pi_k = \pi_0 P^k.$$

Eigenvalues of P determine convergence to stationarity. If the graph is connected and non-bipartite, the walk converges to the uniform distribution.

The same object has two readings: A^k counts deterministic walks, while P^k describes probabilities. These finite graph-walk calculations are the beginning of Markov-chain theory.

§1.19 Catalan recursion and parenthesization

Polygon triangulations and parenthesizations are the same combinatorial structure. A triangulation of an $(n+2)$-gon corresponds to a way of multiplying $n+1$ factors with binary parentheses. Choosing the triangle incident to a fixed edge splits the polygon into two smaller polygons, producing the Catalan recursion.

This is the combinatorics behind the associahedron, a polytope whose vertices are parenthesizations and whose edges are single reassociations:

$$(ab)c \leftrightarrow a(bc).$$

The elementary Catalan count therefore touches the geometry of moduli spaces of bracketings.

§1.20 Adjacency spectra and expansion

The theorem that $(A^k)_{ij}$ counts walks becomes much stronger when combined with eigenvalues. If a graph is d-regular with adjacency eigenvalues

$$d = \lambda_1 \geq |\lambda_2| \geq \cdots ,$$

then $|\lambda_2|$ controls how quickly random walks mix and how well the graph expands.

Expander graphs are sparse graphs that behave like highly connected graphs. They are central in theoretical computer science, coding theory, and combinatorics. The elementary walk-counting matrix is the first tool for studying them.